

3-Mark Questions & Answers -- Data Structures Module 2: Stacks & Queues

Q1. Explain the algorithm for checking Well-Formedness of Parentheses using a Stack.

1. Scan the arithmetic string expression from left to right.
2. If a

Q2. Differentiate between Infix, Prefix, and Postfix notations.

Notation	Operator Position	Example	Evaluation
Infix	Operator between operands.	`(A + B) * C`	Requires operator precedence and paren
Prefix (Polish)	Operator before operands.	`* + A B C`	Evaluated from right to left without p
Postfix (Reverse Polish)	Operator after operands.	`A B + C *`	Evaluated from left to right using a S

Q3. Explain Priority Queue and its types.

A Priority Queue is a queue where each element has an assigned priority:

* As